

Effective Load: Distinction Between External and Internal Load in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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April 15, 2026 — Version 1

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Abstract

This document defines the distinction between external load and effective load in Dimensional Human Field Theory (DHFT).

External load is total demand present in the environment.

Effective load is external load that enters and acts on the system.

I. Scope

This document establishes the distinction between external load and effective load within DHFT.

Effective load is defined as a structural condition of system interaction with environmental demand.

No additional variables are introduced.

II. Definition

External load is total demand present in the environment independent of the system.

Load (L) is demand acting on the system.

Effective load is the portion of external load that enters and acts on the system under boundary conditions.

III. Distinction

External load and effective load are not equivalent.

The magnitude of external load does not determine the magnitude of load acting on the system.

Load acting on the system is determined by boundary conditions governing entry, containment, and transfer.

IV. Boundary Relation

Effective load is constrained by boundary conditions.

Boundary conditions determine whether external load:

- enters the system
- is excluded from the system
- is partially transmitted

V. Non-Equivalence Condition

Identical external load produces different effective load under differing boundary conditions.

System state is determined by effective load relative to capacity, not by external load alone.

VI. Constraint

All valid system descriptions distinguish between external load and load acting on the system.

No additional variables are introduced. Effective load is defined within the existing invariant structure.

VII. Statement

Effective load is the portion of external load that enters and acts on a system under boundary conditions.

External load and effective load are not equivalent.

VIII. System Reference

The system defined in this document is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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